

Well-Founded Expansion and Lexicographic Well-Ordering for 1-Y

A proof exposition with a Lean 4 formalization

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Abstract

The 1-Y system expands finite sequences by constructing difference mountains, extracting column tops with their inherited ancestor relations, and copying blocks across several extraction layers. We explain a proof that every expansion trajectory terminates, for arbitrary choices of finite copy counts, and that the standard generated sequences are well-ordered lexicographically. The proof associates an ordinal representation to the entire finite parent diagram. Its edge conditions use root-indexed relations defined by Σ_1 preservation between constructible levels equipped with recursively constructed truth predicates. A finite reflection lemma allows copied blocks to be represented below the previous last-column label. Ordinal induction then proves well-foundedness; additional prefix and reachability arguments give lexicographic well-ordering. The complete Lean 4 formalization has passed kernel verification. We also present the accompanying set-bounded ZFC argument. Its complete formal migration to a first-order ZFC derivation has not yet been machine-verified.

Formalization repository

<https://github.com/Phyrion1343/1Y-Well-Ordering-Lean>

Verified source snapshot: `df3b69eea954c4640067e2b79d6c238a7217ce24`

1 Introduction and statement of results

The connection between 0-Y and 1-Y suggests a termination argument: a one-layer 1-Y expansion follows the corresponding 0-Y rule, and a multi-layer expansion is controlled by a terminal extraction layer with the same kind of rule. The ancestor relation is part of this correspondence. Replacing an extracted object by its numerical sequence alone changes the construction.

The lower layers also matter. Contour lifting and reference filling can increase the physical heights of columns and create new parent edges. Thus a proof must relate the terminal-layer operation to every reconstructed lower layer and to subsequent expansions. The method developed here records all of these edges in a finite diagram and equips that diagram with ordinal labels. This is in the broader tradition of representation arguments for sequence systems, including the well-ordering proof for the Bashicu Matrix System [1]. The present exposition describes the concrete 1-Y formalization [2]; it does not assume that a numerical projection to 0-Y is a decreasing rank.

The mountain and extraction terminology is also used in the community description of Y-sequences [5]. The precise convention considered here is the ancestor-preserving algorithm

below and in the fixed source snapshot. No equivalence with every variant described elsewhere is assumed.

Let

$$\mathcal{E} = \{\emptyset\} \cup \{(a_0, \dots, a_{m-1}) : m > 0, a_0 = 1, a_i \in \mathbb{N}_{>0}\}.$$

Write $E_N(s)$ for the actual ancestor-preserving 1-Y expansion with N *additional* copies of the bad block. In particular, $E_0(s)$ deletes the last entry, and $E_N(\emptyset) = \emptyset$. The numerical function in the formalization is `OneY.Numeric.expand`; the name `ZeroY.Expr` is reused only for the common carrier of finite sequences.

Definition 1.1. The nontrivial expansion relation is

$$t \prec s \iff t \neq s \text{ and } (\exists N \in \mathbb{N}) t = E_N(s).$$

Let $\text{Desc}(s)$ consist of all sequences reachable from s by finitely many expansions, including s itself. Let $\mathcal{G}$ consist of the sequences reachable from a seed $(1, m)$ with $m \geq 2$. Lexicographic order uses the convention that a proper prefix is smaller.

Theorem 1.2 (Main theorem). *For the 1-Y expansion specified in Section 2:*

- (i) $\prec$ is well-founded on $\mathcal{E}$.
- (ii) If $s_{n+1} = E_{N_n}(s_n)$ for arbitrary $N_n \in \mathbb{N}$, then $s_n = \emptyset$ for some n .
- (iii) For every $s \in \mathcal{E}$, the lexicographic order on $\text{Desc}(s)$ is a strict well-order.
- (iv) The lexicographic order on $\mathcal{G}$ is a strict well-order.

These four conclusions are the final, kernel-checked Lean theorems listed in Section 11. They require no additional representation, reflection, reconstruction, or well-foundedness hypothesis from their users.

The domains in the theorem are essential. The lexicographic order on all of $\mathcal{E}$ is not well-founded:

$$(1, 2) >_{\text{lex}} (1, 1, 2) >_{\text{lex}} (1, 1, 1, 2) >_{\text{lex}} \dots$$

Each of these sequences nevertheless has terminating expansion trajectories.

Organization and level of detail. We state the finite reconstruction and copying results needed by the proof, explain their invariants, and identify their formal counterparts. Their row-by-row case analyses are summarized as proof sketches; the linked repository supplies the complete checked arguments. The ordinal descent and the passage from expansion to lexicographic order are proved explicitly. Section 10 explains the set-bounded ZFC reconstruction and its verification status. We make no claim to determine the exact order type or the weakest theory proving the theorem.

2 The ancestor-preserving expansion

2.1 Rows, ancestors, and extraction

Number columns from 0. In the initial row a , the parent of column c is

$$P_0(c) = \max\{p < c : a_p < a_c\},$$

when the set is nonempty. Its ancestor chain is obtained by iterating the parent map. Every parent edge points strictly left.

In a difference mountain, a node in row $r + 1$ and column c exists precisely when the node below it has a parent in row r . Its value is

$$v_{r+1}(c) = v_r(c) - v_r(P_r(c)). \quad (2.1)$$

The parent of this new node is selected only among the ancestor positions inherited from row r , taking the nearest position with smaller value. Missing nodes are not numerical zeros. In particular, the convention that the first entry persists in every difference row is not used.

Let $H(c)$ be the highest occupied row of column c in a layer. Its extracted value is $U(c) = v_{H(c)}(c)$. The next layer inherits a forest as well as these values. The geometric intermediate forest Q is characterized as follows: if $H(c) = 0$, then $Q(c)$ is undefined; otherwise follow the ancestor chain in row $H(c) - 1$ and take the nearest candidate p with

$$H(p) \in \{H(c) - 1, H(c)\}.$$

On the resulting Q -ancestor chain, select the nearest entry with value strictly smaller than $U(c)$. This gives the parent relation of the extracted row. This description is part of the implemented extraction rule, rather than a reconstruction from the extracted numbers alone.

We use k for the extraction layer, r for a finite row within that layer, and c for a column. Later, $f(c)$ will be an ordinal label assigned in the proof. None of these indices is a numerical entry of the sequence.

Lemma 2.1 (Finite construction). *Every $s \in \mathcal{E}$ has a finite complete extracted mountain. If an entry at the bottom of a layer exceeds 1, its extracted top value is strictly smaller; an entry equal to 1 remains 1. Consequently $\max(1, \max s)$ is a sufficient bound on the number of layers needed to include a final all-one layer.*

Proof. All values are positive. A non-root entry has a strictly smaller parent, so taking its first difference strictly reduces its value; further existing differences do not increase it. Roots have value 1, a property preserved by extraction with the inherited forest. More precisely, if the value in layer k at a fixed column still exceeds 1, that value plus k is at most the original entry. The claimed bound follows. Each occupied column also has finite height because an upward step strictly decreases its positive value. $\square$

Formalization. [Extraction.lean](#).

This proves that one expansion can be constructed. It does not yet prove termination of an iterated expansion, whose new entries may be much larger.

2.2 The active edge and block copying

For a nonempty input let $x = |s| - 1$. If the last entry has no parent, expansion simply deletes it. Otherwise there is a unique active bad-root configuration (K, d, y) : in extraction layer K and row d , the last column x has parent y , and the corresponding difference is 1. The selection proceeds through the inherited extraction structure until this configuration is reached. Put $\ell = x - y > 0$. The good part consists of columns strictly before y ; the bad block is $[y, x)$.

In the active layer the operation follows the inherited-forest 0-Y copying rule. Parent relations at the deleted last position remain available as boundary data. Other layers are reconstructed using contour lifting, horizontal translation, and reference filling. A right leg connects a node vertically to its difference node; the corresponding left leg connects that difference node to the parent used in subtraction. Contour paths are identified by the alternating-leg rule from the root-column tops. During copying, contour edges receive the

prescribed height increase; translation edges retain their height, and translation parents to the left of the root remain anchored. Translated copies of the reference edges fill the resulting gaps. Heights are computed for the current copy, so the rightmost column is not treated as fixed throughout the operation.

The formal development resolves these geometric instructions into finite parent maps before reconstructing values. This order matters: the resulting numerical sequence must regenerate those same parent maps on its next use.

Lemma 2.2 (Canonical reconstruction). *For the actual upper-layer values produced by expansion, rebuilding the complete mountain from $E_N(s)$ reproduces the prescribed extraction layers, column heights, and parent relations on the finite output width.*

Proof sketch. Reconstruction adds parent values back to the corresponding differences. It is necessary to prove both that each intended parent is smaller and that every closer inherited candidate is blocked. The proof establishes these comparison and blocking conditions for the actual expanded top values, then proceeds through the active layer and the layers below and above it. Prefix compatibility limits each assertion to the actual output width. Redundant all-one layers introduce no further parent edges. Arbitrary assignments of top values would not satisfy this argument. $\square$

Formalization. [ExpansionCanonical.lean](#).

Example 2.3. The tops extracted from $(1, 3, 3)$ are $(1, 2, 2)$. One additional copy gives the extracted row $(1, 2, 1, 2)$ and the actual bottom row

$$E_1(1, 3, 3) = (1, 3, 2, 5).$$

The occupied nodes in the first layer of this output are

Finite row	Column 0	Column 1	Column 2	Column 3
2				2
1		2	1	3
0	1	3	2	5

Thus the fourth extracted entry 2 is now at row 2. Its column contains the reference contribution 3 at row 1. An extracted position follows the column top, not a fixed physical height. These values and heights are verified by direct Lean calculations.

Formalization. [ExpansionExamples.lean](#).

3 Finite parent diagrams and their representations

For each layer k and row r , let $q_{k,r}(c)$ be the root reached by following the row's parent map from c . If $P_{k,r}(c) = p$, then

$$q_{k,r}(c) = q_{k,r}(p) \leq p < c.$$

Record this edge by the atom (k, q, p, c) , with $q = q_{k,r}(c)$. The finite diagram $G(s)$ consists of the column set and all such atoms, over the finite complete mountain. Duplicate atoms do not affect its representation conditions.

Definition 3.1. Suppose D is a class of ordinals and $R(k, \eta, a, b)$ is a relation. A representation of a diagram with m columns is a function $f : m \rightarrow \text{Ord}$ such that

$$\begin{aligned} i < j < m &\implies f(i) < f(j), \\ i < m &\implies D(f(i)), \\ (k, q, p, c) \in G &\implies R(k, f(q), f(p), f(c)). \end{aligned}$$

Write $\text{Rep}(G, f)$ for these conditions.

The component root q is not the bad-root column y of an expansion. The component containing the active parent edge can begin strictly to the left of y . This extra index records information that survives changes in physical row height. Copied edges will have either the transported root or an earlier root. The semantic relation will permit weakening the root index in the latter case.

The finite combinatorial argument uses three properties of R : strictness in its endpoints, weakening of its root index, and a finite reflection principle. We now construct the relation and supply these properties mathematically.

Formalization. `RootIndexed/Representation.lean`.

4 Canonical truth structures and ordinal relations

4.1 Set-sized satisfaction and mixed truth towers

For a set-sized relational structure M , its complete first-order satisfaction relation is a set. It can be constructed by recursion on formula depth, using the full set of legal finite assignments. The Tarski clauses determine the result uniquely. We use this ordinary set construction at each stage below.

Fix a set domain U and an ordinal bound a . The stages are $\mathbb{N} \times (a + 1)$, ordered lexicographically with the natural-number coordinate first. At stage (k, η) the language has membership, separate named truth predicates for earlier indices $\xi < \eta$ in the same block, and one diagonal truth predicate for each lower block $j < k$. A diagonal predicate accepts an index code as an argument; a named predicate has its index fixed in the language. The latter distinction controls which parameters can be used in finite reflection.

More explicitly, if $T_{k,\eta}$ denotes the satisfaction set already constructed at stage (k, η) for the fixed domain and bound, then

$$\begin{aligned} S_\xi(e, t) &\iff \langle e, t \rangle \in T_{k,\xi} & (\xi < \eta), \\ D_j(z, e, t) &\iff (\exists \xi < a) [z = \text{code}(\xi) \wedge \langle e, t \rangle \in T_{j,\xi}] & (j < k). \end{aligned}$$

Here e and t are formula and assignment codes. The diagonal index ranges over $\xi < a$, independently of the current same-block index η ; the named predicate has its fixed index $\xi < \eta$.

Each truth predicate reads a strictly earlier stage. Well-founded recursion therefore constructs the full tower, with the current value equal to the complete satisfaction set of the current structure. On $U = L_a$, write the canonical stage structure as $M_{k,\eta}^a$.

All formulas in this discussion are finite. A formula may belong to a language with many named predicates, but it uses only finitely many of them. The formalization proves finite-support and language-restriction lemmas before using this observation.

Formalization. `OneYTruth/ExternalTower.lean`.

4.2 The root-indexed relation

Bounded quantifiers mean set-theoretic quantifiers of the form $\forall x \in z$ or $\exists x \in z$. In each expanded language, Δ_0 formulas use only bounded quantifiers, and the Σ_1 fragment used here consists of a finite existential prefix followed by a Δ_0 matrix. This is a syntactic fragment; finite variable scope alone does not make a formula Δ_0 .

Definition 4.1. Let $R(k, \eta, a, b)$ mean that $\eta \leq a < b$ and the natural inclusion

$$M_{k,\eta}^a \longrightarrow M_{k,\eta}^b$$

preserves and reflects the truth of every formula in this Σ_1 fragment, with parameters from the smaller domain.

In particular, an existential assertion true in the larger structure can be reflected to the smaller structure, provided its formula belongs to the available language and its parameters lie in L_a .

Lemma 4.2. *The canonical relation satisfies*

$$\begin{aligned} R(k, \eta, a, b) &\implies a < b, \\ \xi \leq \eta \text{ and } R(k, \eta, a, b) &\implies R(k, \xi, a, b), \\ R(k, \eta, a, b) \text{ and } R(k, \eta, b, c) &\implies R(k, \eta, a, c). \end{aligned}$$

Moreover, if $k < K$, $\eta \leq a$, and $R(K, \theta, a, b)$, then $R(k, \eta, a, b)$.

Proof. Strictness is a defining conjunct. Index weakening follows by restricting the set of named predicates. Transitivity follows by composing the two inclusions and their truth equivalences. For the final assertion, translate the lower-block language into the higher-block language using its diagonal predicate and the relevant index codes. The canonical tower equations identify the resulting reduct with the required lower-block structure. $\square$

Definition 4.3. An ordinal a is *adequate* if $\omega < a$, a is a nonzero limit, and every $M_{k,\eta}^a$, for $k < \omega$ and $\eta \leq a$, satisfies the Separation and Collection schemes in its expanded language. Set $D(a) = \text{Adequate}(a)$.

Adequacy provides the weak internal closure needed for the proof. It does not include Power Set and does not assert that L_a is a transitive model of all of ZFC. Full truth sets for the current ambient domain are not assumed to be elements of that same domain.

Formalization. `OneYTruth/RootSemantics.lean`.

5 Initial representations

Set $\kappa = \omega_1^V$ and $U = L_\kappa$. The superscript indicates the first uncountable ordinal of the ambient universe; no equality with ω_1^L is needed. All members of U are externally countable. Let W encode the canonical truth information on U for all finite blocks and indices below κ . The auxiliary language is finite: it has membership and a single four-place predicate for W .

Lemma 5.1 (Constructible set satisfaction). *If a set domain and its set-sized relation tables are constructible, their complete finite-formula satisfaction set is constructible and agrees with the external satisfaction relation. The canonical set-length truth towers and the auxiliary table W are also constructible.*

Proof outline. Construct the full finite syntax and assignment sources, and compute satisfaction by recursion on formula depth. Each step reads atomic relation tables and earlier truth values; quantified variables range over the fixed set domain. Separation performs a step, Replacement collects the finite stages, and union collects their values. The fixed construction and its uniqueness proof can be relativized to L . Finite syntax and assignments have the same codes inside and outside L , and the Tarski clauses then identify the two solutions.

For the mixed tower, apply set-length well-founded recursion to the fixed step operator whose value is this set satisfaction. Carry out the same recursion inside L , and use uniqueness on the stage order to identify the result. The required table is obtained by fixed definable collection and projection from the tower. This uses set domains throughout; it does not construct a uniform truth predicate for the proper class L . $\square$

Lemma 5.2 (Constructible subsets below ω_1). *If $a \in L_\kappa$, $b \subseteq a$, and $b \in L$, then $b \in L_\kappa$.*

Proof. Choose a countable constructible stage L_γ containing a and hence every element of a . In a sufficiently large limit stage L_δ , take a countable elementary substructure containing b and all elements of the transitive seed L_γ . By condensation its transitive collapse is L_β for some $\beta < \kappa$. The collapse fixes L_γ pointwise and therefore fixes b , since every member of b lies in the seed and is present in the substructure. Hence $b \in L_\beta \subseteq L_\kappa$. $\square$

Lemma 5.3 (The initial auxiliary structure). *The structure $\mathcal{A} = (U, \in, W)$ satisfies Separation and Collection in its auxiliary language.*

Proof. For Separation, Lemma 5.1 gives a constructible set $S = \text{Sat}(\mathcal{A})$. A formula code e , parameters p , and $a \in U$ determine

$$b = \{x \in a : \langle e, p \frown x \rangle \in S\}.$$

This is a fixed set-theoretic Separation instance with code parameters, so $b \in L$. Lemma 5.2 places b back in U .

For Collection, an internal input set $a \in U$ is externally countable. Choose one witness in U for each of its elements. The countably many constructible stages of these witnesses have a common bound below κ . A sufficiently large $L_\beta \in U$ then contains all the witnesses. This argument uses choice for a set-indexed family. $\square$

Lemma 5.4 (Club supply). *There is a closed unbounded set $C \subseteq \kappa$ of limit ordinals above ω such that every $\alpha \in C$ is adequate and*

$$\alpha < \beta \text{ in } C, \quad \eta \leq \alpha \implies R(k, \eta, \alpha, \beta) \quad (k < \omega).$$

Proof outline. In the finite language of $\mathcal{A}$, alternate countable Skolem closure with enlargement to complete countable constructible stages. The union of an ω -chain of these closures is an elementary substructure of the form $\mathcal{A} \upharpoonright L_\alpha$. This proves unboundedness of the set of such stages. The elementary chain argument proves closure.

To transfer the auxiliary information to the canonical mixed structures, induct on $\omega \times (\alpha + 1)$. At a stage, previously established trace equalities handle same-block named predicates and lower-block diagonal predicates. Translate the current formula to the auxiliary language and use elementarity. Its named indices are finite in number and strictly below the relevant bound. Even when $\eta = \alpha$, each particular formula uses only indices below α ; α itself need not be an internal parameter. This proves agreement between the restricted large tower and the tower recomputed over L_α .

Separation and Collection transfer from Lemma 5.3 by elementarity and the canonical trace identifications, giving adequacy. Comparing the two traces for $\alpha < \beta$ gives the asserted R relation. $\square$

Formalization. `OneYTruth/InitialCanonicalSupply.lean`.

Corollary 5.5 (Initial representation). *Every finite diagram whose atoms satisfy $q \leq p < c$ has a representation. In particular, every $G(s)$ has one.*

Proof. For a diagram of width m , choose m increasing members of C as labels. They are adequate. For an atom (k, q, p, c) , we have $f(q) \leq f(p) < f(c)$, and Lemma 5.4 supplies $R(k, f(q), f(p), f(c))$. $\square$

6 Finite reflection with endpoint requirements

The next lemma reflects an entire finite representation at once. Its requirements distinguish an internal edge from an edge whose upper endpoint is the current ambient domain. This distinction prevents the argument from treating an ambient ordinal as one of its own elements.

Definition 6.1. For a representation f , an ambient upper endpoint β , a cut column c , and a control relation $R(K, \theta, f(c), \beta)$, an endpoint requirement is a triple (k, q, p) asserting $R(k, f(q), f(p), \beta)$. It is *admissible* if either

$$k < K, \quad \text{or} \quad k = K, \quad q < c, \quad f(q) < \theta. \quad (6.1)$$

All its referenced columns are in the finite diagram and satisfy $q \leq p$.

Theorem 6.2 (Finite reflection). *Let G have width m and representation f . Suppose $c < m$, $\alpha = f(c)$, every $f(i) < \beta$, $\text{Adequate}(\beta)$, and $R(K, \theta, \alpha, \beta)$. Let $\mathcal{N}$ be a finite family of admissible endpoint requirements, all true at f and β . Then there is a representation g of G such that*

$$g(i) = f(i) \quad (i < c), \quad g(i) < \alpha \quad (i < m), \quad (6.2)$$

and, for every $(k, q, p) \in \mathcal{N}$,

$$R(k, g(q), g(p), \alpha). \quad (6.3)$$

Proof sketch; full certificate details are formalized. The proof encodes the existence of g as one Σ_1 formula of the control language. Its fixed parameters are ω , the empty set, the finitely many layer numbers used, and the labels before the cut. All are elements of L_α .

The formula existentially quantifies the other column labels and certificates for the following assertions: the labels are strictly increasing ordinals; their domains are the corresponding L -stages; they are adequate; all internal diagram relations hold; and the listed endpoint requirements hold. The certificates include full syntax, assignment and stage sources, canonical satisfaction sets, complete recursion histories, and the data needed to check the Separation and Collection schemes. Coverage and uniqueness clauses identify the intended objects. Checking only a self-consistent arbitrary subcollection would not suffice.

These certificates exist inside L_β . Adequacy supplies Separation and Collection, and hence the fixed instances of Replacement needed for set-length recursion. The stage source, its well-order, and its initial segments are uniformly definable by fixed formulas with parameters in the domain. Merely knowing that each individual predecessor set belongs to the domain would not suffice. Local recursion solutions are collected and united; their uniqueness ensures compatibility. For the remaining locally finite witnesses, Collection supplies one common set

bound. After these witnesses are existentially quantified, all checks are bounded, giving the required literal Σ_1 form. No Power Set axiom for L_β is used.

Schematically the common-bound step has the form

$$\exists g \exists B [\text{bounded source and graph checks} \wedge \forall s \in S \exists w \in B \Psi(s, w, g)],$$

where S is the complete stage source and Ψ is Δ_0 . The quantified witnesses include the needed source certificates. This pattern explains why local existence plus Collection yields a finite existential prefix, while still checking every stage in the intended source.

An endpoint requirement cannot be handled by asking for the complete truth set of L_β as an element of L_β . If $k < K$, its formula instead uses a lower-block diagonal predicate. In the second case of (6.1), its root index is a named predicate available in the control language, and that index is among the fixed prefix parameters. Internal diagram edges have two endpoints strictly below the ambient domain; their certificates may involve any of the finitely occurring blocks, with no restriction $k \leq K$.

The old representation f supplies a witness to the complete formula in L_β . The control relation reflects this same formula to L_α . Decode its witness there. First check the ordinal and order assertions; then adequacy and the inequalities $q \leq p$ provide the guards required by Definition 4.1. Certificate soundness yields all internal relations, and the endpoint formulas yield (6.3). The prefix equalities and domain membership give (6.2). $\square$

The hard internal-existence and syntactic-complexity steps in this proof are separate checked modules, not assumptions of the final result. The formal theorem is `RootSemantics.actual_finiteReflection`. Its hypotheses include the adequacy of β explicitly.

Formalization. `OneYTruth/ActualFiniteReflection.lean`.

7 Copying blocks below the previous last label

Return to a nontrivial bad-root configuration with last column x , bad root y , active block K , and bad-block length $\ell = x - y$. After deleting the last column and making b additional copies, the width and cut are

$$n_b = x + b\ell, \quad c_b = y + b\ell, \quad n_b - c_b = \ell. \quad (7.1)$$

The old last label, denoted β , is retained as a virtual upper endpoint; it need not label an actual output column.

Lemma 7.1 (Finite copying geometry). *The actual copied mountain admits a block-by-block invariant consisting of its current finite diagram, a cut, a control root, saved source-edge facts, and endpoint templates. At each step its new edge atoms are covered by:*

- (i) *old atoms in the reflected part;*
- (ii) *transported source atoms, possibly with an earlier component root;*
- (iii) *seam atoms supplied by an admissible endpoint requirement.*

The control relation and the saved facts and templates needed for the next step are preserved by the prescribed update.

Proof sketch. Compute parent positions and component roots in the three regions: the active layer, layers below it with contour lifting and reference filling, and layers above it. The copied

source relations provide the transported edges; component roots are transported or weakened to earlier columns. The seam relations are classified by extraction block. Nontrivial lower-block requirements use the first alternative of (6.1). A same-control-block seam requirement has its root strictly to the left of the control root, giving the second alternative.

The invariant stores more than the current graph's edge conditions. A later copy can require an original source relation stronger than the weakened relation currently visible at an edge. Saved facts retain these relations, and templates retain requirements pointing to the old virtual endpoint. Their explicit column-translation formulas prove the update clauses. The construction includes the parent data of the deleted last position, which is needed at the seam. $\square$

Formalization. `RootIndexed/ActualScheme.lean`.

Lemma 7.2 (One-block splice). *Assume the invariant of Lemma 7.1, with a representation f of a current graph of width n , a cut c , and all labels below an adequate virtual endpoint β . Then the next copied block can be represented with all its labels still below β .*

Proof. Put $\alpha = f(c)$. The invariant gives the control relation and admissible endpoint requirements for Theorem 6.2. Obtain a reflected representation g with every value below α and with the prefix before c fixed. Append the saved old tail labels:

$$h(i) = \begin{cases} g(i), & i < n, \\ f(c + i - n), & n \leq i < n + (n - c). \end{cases} \quad (7.2)$$

The first part is strictly below α , and the appended part starts at $f(c) = \alpha$. Both parts are strictly increasing, so h is strictly increasing. All its labels are adequate and below β .

For an old atom use the reflected representation. For a transported atom use a saved source fact and, where needed, index weakening from Lemma 4.2. For a seam atom use the reflected endpoint requirement with new endpoint α . These cover the actual new graph by Lemma 7.1. The appended block retains the relevant old labels, so the saved facts and templates update as required for the next step. This proves the assertion together with the full induction invariant. $\square$

Theorem 7.3 (Last-label descent). *Suppose $s \neq \emptyset$ and $\text{Rep}(G(s), f)$, with last label $\beta = f(|s| - 1)$. For every $N \in \mathbb{N}$, if $E_N(s) \neq \emptyset$, there is a representation of $G(E_N(s))$ whose last label is less than β .*

Proof. If the last entry has no parent, the output diagram is the corresponding proper prefix, so restrict f . Its last label is smaller by strict increase. The same restriction handles zero additional copies.

In the copying case, initialize the diagram invariant from $G(s)$ while deleting its last column. The virtual endpoint β is adequate because it was a column label. All remaining labels are below β ; the active parent edge supplies the initial control relation, and the original edges supply the initial facts and endpoint templates. Induct on N using Lemma 7.2 and (7.1). The final copied diagram has all labels below β . Canonical reconstruction (Lemma 2.2) identifies its finite parent diagram with that of the actual output $E_N(s)$. Restriction to the actual output width therefore gives the required representation. $\square$

8 Well-foundedness and arbitrary-strategy termination

For nonempty s , let $\text{LastRep}(s, \beta)$ mean that $G(s)$ has a representation with last label β . The descent theorem gives

$$\text{LastRep}(s, \beta), \quad E_N(s) \neq \emptyset \implies (\exists \gamma < \beta) \text{LastRep}(E_N(s), \gamma). \quad (8.1)$$

Representations need not be unique or computable. The induction below allows them to change at every expansion step.

Proof of Theorem 1.2(i)–(ii). For a relation written in the downward direction $t \prec s$, accessibility of s means that every $t \prec s$ is accessible. By ordinal well-founded induction on β , prove

$$\forall s \neq \emptyset \quad \text{LastRep}(s, \beta) \implies \text{Acc}_{\prec}(s).$$

If $t \prec s$ is empty, it is accessible because it has no nontrivial expansion. Otherwise $t = E_N(s)$ for some N , and (8.1) supplies a smaller label γ representing t . Apply the induction hypothesis.

Corollary 5.5 supplies a representation of every nonempty input. The empty input is already accessible, so $\prec$ is well-founded on all of $\mathcal{E}$.

Every expansion of a nonempty input is lexicographically smaller (Lemma 9.1 below), hence nontrivial. A trajectory avoiding the empty sequence would therefore be an infinite $\prec$ -descending chain, contradicting accessibility of its initial sequence. Thus it reaches the empty sequence. No effectiveness or monotonicity condition on the copy counts is needed. $\square$

Formalization. `RootIndexed/ExpansionWellFounded.lean`.

9 From expansion to lexicographic well-ordering

Well-founded expansion alone does not identify the order of arbitrary sequences. The following numerical properties connect it to the generated lexicographic orders.

Lemma 9.1 (Numerical and prefix properties). *For $s \neq \emptyset$ and $N \in \mathbb{N}$:*

- (i) $E_N(s) <_{\text{lex}} s$.
- (ii) If $i \leq j$, then $E_i(s)$ is a prefix of $E_j(s)$.
- (iii) Every prefix of a sequence is reachable from it by repeated zero-copy expansions.

Proof and implementation-dependent part. For the deleting cases, (i) follows from the proper-prefix convention. In the positive-copy bad-root case, the finite numerical reconstruction preserves all entries before the former last column x , and the first seam entry is exactly $s[x] - 1$. This is the first differing position, proving (i), independently of the length or size of the remaining tail. For (ii), the copying and reconstruction formulas agree on the width of the shorter output. This prefix equality is established for the complete canonical expansion. Finally, E_0 deletes precisely one last entry, which proves (iii). $\square$

Formalization. `ExpansionOrder.lean`.

Write $a \preceq_* b$ when a is reachable from b , and $a \prec_+ b$ when a nonempty path of nontrivial expansions leads from b to a .

Lemma 9.2 (Comparability of descendants). *For each s , any two elements of $\text{Desc}(s)$ are comparable by $\preceq_*$.*

Proof. Use well-founded induction on s under $\prec$. The assertion is immediate if one of the two descendants is s . Otherwise their paths begin with $E_i(s)$ and $E_j(s)$. Assume $i \leq j$. By Lemma 9.1, $E_i(s)$ is a prefix, and hence a descendant, of $E_j(s)$. Thus both chosen sequences are descendants of $E_j(s)$. For a nonempty s , this immediate successor is strictly below s , so the induction hypothesis applies. The empty case has a singleton descendant set. $\square$

Proposition 9.3. *If $a, b \in \text{Desc}(s)$, then*

$$a <_{\text{lex}} b \iff a \prec_+ b. \quad (9.1)$$

Proof. Every nontrivial expansion strictly decreases lexicographic order, proving the right-to-left direction. For the converse use Lemma 9.2. Equality is excluded. If a nontrivial path instead led from a to b , its lexicographic decrease would contradict $a <_{\text{lex}} b$. Hence the path has the required direction. $\square$

Proof of Theorem 1.2(iii)–(iv). The transitive closure of a well-founded relation is well-founded. By (9.1), lexicographic order on $\text{Desc}(s)$ is its restriction there. Transitivity and trichotomy are inherited from finite-sequence lexicographic order, giving (iii).

Use the convenient seed notation $\text{seed}(n) = (1, n + 1)$. The actual expansion satisfies

$$E_1(\text{seed}(n + 1)) = \text{seed}(n).$$

Thus any two seeds, and consequently any two standard generated sequences, have a common seed ancestor. Proposition 9.3 applies to each pair in $\mathcal{G}$, so lexicographic order on $\mathcal{G}$ coincides with the restriction of $\prec_+$. This proves (iv). Allowing the degenerate seed $(1, 1)$ would not enlarge $\mathcal{G}$, since it is already generated from $(1, 2)$. $\square$

Formalization. [OneY/Dynamics.lean](#).

10 A set-bounded ZFC reconstruction

The preceding representation method also admits the following ordinary set-theoretic presentation. This section identifies a single set containing all labels and all semantic data used in the argument. Its purpose is to explain the accompanying ZFC proof, separately from the foundation in which the existing Lean proof is checked.

10.1 A common set of semantic data

Keep $\kappa = \omega_1^Y$. Set recursion constructs $\langle L_a : a \leq \kappa \rangle$. For each $a \leq \kappa$, set-length recursion on $\omega \times (a + 1)$ gives the full canonical tower on L_a . Replacement collects these towers into a single set $\mathcal{T}_\kappa$. Formula codes and finite assignments are set elements, so R is a subset of $\omega \times \kappa^3$, and adequacy is a subset of κ .

Constructibility of the towers follows from the fixed set recursion in Lemma 5.1. In particular, the variable input formula is a code parameter in a fixed satisfaction query. We do not quantify over all formulas while invoking a uniform class truth predicate for L . The semantic presentation of a particular fixed formula relativized to L must be distinguished from such a class truth predicate.

Within an adequate L_β , the certificate argument of Theorem 6.2 uses set-length recursion, Separation, Collection, finite set operations and ω . All witnesses are for smaller candidate domains. Reflection produces smaller labels, so it also respects any further fixed upper bound on the labels.

10.2 One countable ordinal bounds the construction

Theorem 10.1 (A countable ranking function). *In the set-theoretic argument there are a countable ordinal ρ and a function $\mu : \mathcal{E} \rightarrow \rho$ such that*

$$s \neq \emptyset \implies \mu(E_N(s)) < \mu(s) \quad (N \in \mathbb{N}). \quad (10.1)$$

Moreover, μ restricted to $\mathcal{G}$, or to any fixed $\text{Desc}(s)$, is strictly increasing for lexicographic order.

Proof. Choose an increasing sequence $\alpha_0 < \alpha_1 < \dots$ in the club C of Lemma 5.4, and set

$$\rho = \sup_{n < \omega} (\alpha_n + 1) < \omega_1.$$

For a finite diagram of width m , the assignment $f(i) = \alpha_i$, $i < m$, gives a representation below ρ . Hence every input has such a representation using the same countable label bound. Reflection and splicing only lower the old bound, so Last-label Descent remains valid when all representations are restricted to ρ .

For nonempty s , form the nonempty set

$$B_s = \{b < \rho : (\exists f \in \rho^{|s|}) [\text{Rep}(G(s), f) \wedge f(|s| - 1) = b]\}.$$

Define

$$\mu(\emptyset) = 0, \quad \mu(s) = \min B_s \quad (s \neq \emptyset).$$

Separation defines B_s within an explicit set, and Replacement collects the function μ . This construction uses neither a minimum over a proper class nor global choice.

Choose a representation attaining $\mu(s)$. If $E_N(s)$ is nonempty, Theorem 7.3 supplies a new last label $b' < \mu(s)$, still below ρ . Therefore $\mu(E_N(s)) \leq b' < \mu(s)$. If the output is empty, its rank is 0, while every nonempty last label is adequate and exceeds ω . This proves (10.1).

On a fixed descendant set, or on $\mathcal{G}$, Proposition 9.3 identifies a strict lexicographic comparison with a finite nontrivial expansion path. Repeated application of (10.1) gives the asserted strict increase of μ . $\square$

This provides another proof of the well-ordering conclusions: a nonempty set of states has a state of minimum rank, and no smaller state in the relevant relation can remain in that set. The ranking function need not be computationally useful and does not identify the exact order type.

10.3 What has been formally verified

The 1-Y well-foundedness and well-ordering proofs are complete in Lean 4's dependent type theory and have passed its kernel checker. The accompanying ZFC argument above supplies the mathematical set-theoretic justification: set satisfaction, fixed-formula relativization to L , set recursion, condensation, countable Skolem closure, finite reflection, and a set-valued rank.

The complete migration of this argument into a first-order ZFC proof calculus, including the encoding of the target sentence and the full derivation, has not yet been machine-verified. The standard Lean axiom audit is not a substitute for that migration. These are distinct verification statements; the latter work remaining does not turn the completed Lean well-ordering theorem into an assumed lemma.

11 Formalization and reproducibility

The formalization is publicly available at <https://github.com/Phyrion1343/1Y-Well-Ordering-Lean>. This exposition refers to the immutable snapshot `df3b69e` (the full identifier is given on the first page). The final declarations, all in the namespace `OneYTruth.WellOrdering`, are:

Declaration	Conclusion
<code>expansion_wellFounded</code>	Well-foundedness of actual nontrivial expansion
<code>expansion_chain_reaches_empty</code>	Termination for arbitrary finite copy counts
<code>descendants_strictWellOrder</code>	Lexicographic well-order of each fixed descendant set
<code>generated_strictWellOrder</code>	Lexicographic well-order of the standard generated set

They appear in `OneYTruth/ActualWellOrdering.lean`. The build uses Lean 4.33.1. The complete 1-Y target passed a build of 1814 tasks; the later combined 0-Y and 1-Y release check passed 2015 tasks using existing local build products. Task counts are not counts of new proofs or of freshly recompiled modules. The release audit checks 153 distinct 1-Y declarations, including the four final results and the actual finite reflection theorem. Their axiom dependencies are subsets of `propext`, `Classical.choice`, and `Quot.sound`; the audit excludes `sorryAx` and additional axioms.

The primary framework is Lean [4]; the model-theoretic library is `mathlib` [6]. The repository fixes its dependency revisions and uses the constructible-universe bridge in the BMS formalization [3]. Its two recorded upstream proof-script patches are performance changes with unchanged public theorem statements and assumptions.

On Windows with PowerShell 7, Git, `curl.exe`, and `tar.exe`, the documented commands from the repository root are:

```
./formalization/prepare-toolchain.ps1
./formalization/prepare-dependencies.ps1
./formalization/build.ps1
./formalization/Concrete/build.ps1 -Target OneYTruth
```

The final command builds the target and checks the axiom audit. Dependency sources and build caches are restored separately and are not part of the published source files. The validation record describes the completed checks; the release verification was incremental and did not repeat a new download-and-build cycle from an empty machine.

12 Concluding perspective

The relation to 0-Y appears locally in the control layer and its inherited parent relations. The global argument works because the complete copied mountain has a finite representation invariant. Root indices allow the semantic constraints to follow edges through contour lifting and reference filling; finite reflection then moves the current representation below a cut label before the saved block is appended. The last label decreases after every nonempty expansion.

Two further ingredients finish the theorem: initial representations exist for every finite input, and actual numerical expansions have the prefix properties that identify descendant reachability with lexicographic order. Together they establish the well-foundedness and well-ordering assertions without computing a canonical ordinal notation for each sequence.

Acknowledgments and AI assistance

The author acknowledges substantial assistance from Codex, an AI assistant from OpenAI, throughout this project. Under the author's direction, Codex helped translate the expansion conventions into formal definitions, develop and analyze the proof strategy, implement and debug the Lean formalization, run build and audit checks, and prepare the repository documentation. Codex also drafted and typeset this manuscript. This assistance extended well beyond language editing to mathematical exploration and formalization work. The author supplied the problem and expansion conventions, directed the investigation, and provided reference implementations, examples, and corrections.

A Explicit parent transport in a copied mountain

This appendix gives source-coordinate formulas for the parent maps used in Lemma 7.1. It also specifies how the deleted last position contributes seam data. All formulas concern occupied output columns; an undefined source parent remains undefined after transport.

Let (K, d, y) be the active configuration, x the old last column, and $\ell = x - y$. In the active and lower layers, write the non-prefix source coordinates as

$$c = t + b\ell, \quad y < t \leq x, \quad b \geq 0,$$

and define the anchored parent transport

$$\tau_b(p) = \begin{cases} p, & p < y, \\ p + b\ell, & p \geq y. \end{cases}$$

The case $t = x$ is a seam, since $x + b\ell = y + (b + 1)\ell$. Columns at or before y retain their original data.

In the active layer the height is

$$H'_K(t + b\ell) = \begin{cases} H_K(y), & t = x, \\ H_K(t), & t < x, \end{cases}$$

and the parent map is

$$P'_{K,r}(t + b\ell) = \begin{cases} P_{K,r}(y), & t = x \text{ and } r \geq d, \\ \tau_b(P_{K,r}(t)), & \text{otherwise.} \end{cases}$$

The first branch has its parent to the left of y , when defined.

For a lower layer $k < K$, put

$$h = H_k(y), \quad \delta = H_k(x) - h > 0,$$

$$C_k = \{c : H_k(c) \geq h \text{ and } \text{root}_{P_{k,h}}(c) = y\}.$$

The geometric transport theorem establishes $\text{root}_{P_{k,h}}(x) = y$ and the strict positivity of δ . Heights are transported by

$$H'_k(t + b\ell) = H_k(t) + b\delta \mathbf{1}_{C_k}(t).$$

The complete lower-layer parent formula is

$$P'_{k,r}(t + b\ell) = \begin{cases} P_{k,h}(t) + b\ell, & t \in C_k, \ h \leq r < h + b\delta, \\ P_{k,r-b\delta}(t) + b\ell, & t \in C_k, \ r \geq h + b\delta, \\ \tau_b(P_{k,r}(t)), & \text{otherwise.} \end{cases}$$

The three branches insert reference rows, lift the contour, and translate the remaining edges, respectively. Layers $k > K$ use ordinary sources in $[y, x)$, copying heights and parents within a block and anchoring parents in the good prefix.

After the top values are supplied by the higher-layer expansion, values below a column top are reconstructed by

$$v'_{k,r}(c) = v'_{k,r+1}(c) + v'_{k,r}(P'_{k,r}(c)).$$

These are prescribed finite parent maps. Their agreement with the parent maps recomputed from the final numerical sequence is the additional canonical-reconstruction theorem, not a consequence of defining this sum alone.

Formalization. [OneY/LowerCopy.lean](#).

Formalization. [OneY/TerminalCopy.lean](#).

Formalization. [OneY/OrdinaryCopy.lean](#).

B A map from the proof to the source

All links below refer to the fixed snapshot stated on the first page. Core paths below are under [formalization/](#); [OneYTruth/](#) paths are under [formalization/Concrete/](#). The source contains the full case analyses summarized in this exposition.

Proof component	Source modules
Finite extraction	OneY/Extraction.lean
Canonical numerical rebuilding	OneY/ExpansionCanonical.lean
Root-indexed diagrams	OneY/RootIndexed/Diagram.lean
Representation interface	OneY/RootIndexed/Representation.lean
Copy and splice invariant	OneY/RootIndexed/ActualScheme.lean
Truth towers and semantic relation	OneYTruth/ExternalTower.lean ; OneYTruth/RootSemantics.lean
Initial representation supply	OneYTruth/InitialRepresentations.lean
Adequacy and finite diagram certificates	OneYTruth/ActualAdequateQuery.lean ; OneYTruth/ActualDiagramQueries.lean
Finite reflection	OneYTruth/ActualFiniteReflection.lean
Ordinal descent and accessibility	OneY/RootIndexed/ExpansionWellFounded.lean
Lexicographic order and reachability	OneY/ExpansionOrder.lean ; OneY/Dynamics.lean
Final theorem entry point	OneYTruth/ActualWellOrdering.lean
Audit and build records	Concrete/OneYTruthAudit.lean ; VALIDATION.md

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